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SOLVENCY RATIOS

Learning Outcome

- calculate and interpret activity, liquidity, solvency, and profitability ratios

Solvency refers to a company's ability to fulfill its long-term debt obligations. Assessment of a company's ability to pay its long-term obligations (i.e., to make interest and principal payments) generally includes an in-depth analysis of the components of its financial structure. Solvency ratios, introduced in earlier modules in Corporate Issuers, provide information regarding the relative amount of debt in the company's capital structure and the adequacy of earnings and cash flow to cover interest expenses and other fixed charges (such as lease payments) as they come due.

By analyzing financial statements, an analyst aims to understand levels and trends in a company's use of financial leverage in relation to past practices and the practices of peer companies. Analysts also need to be aware that the greater a company's operating leverage, the greater the risk of the operating income stream available to cover debt payments; operating leverage can thus limit a company's capacity to use financial leverage.

A company's relative solvency is fundamental to valuation of its debt securities and its creditworthiness. Understanding a company's use of debt can provide analysts with insight into the company's future business prospects because management's decisions about financing may signal their beliefs about a company's future. For example, the issuance of long-term debt to repurchase common shares may indicate that management believes the market is underestimating the company's prospects and that the shares are undervalued.

Calculation of Solvency Ratios

The two primary types of solvency ratios are debt ratios and coverage ratios. Debt ratios focus on the balance sheet and measure the amount of debt capital relative to equity capital. Coverage ratios focus on the income statement and measure the ability of a company to cover its debt payments. These ratios are useful in assessing a company's solvency and, therefore, in evaluating the quality of a company's bonds and other debt obligations.

Exhibit 25 describes commonly used solvency ratios. The first three of the debt ratios presented use total debt in the numerator. The definition of total debt used in these ratios varies, with some market participants using the sum of interest-bearing short-term and long-term debt, excluding liabilities such as accrued expenses, accounts payable, and leases. (For calculations in this module, we use this definition.) Other market participants use definitions that are more inclusive (e.g., all liabilities) or restrictive (e.g., long-term debt only, in which case the ratio is sometimes qualified as "long-term," as in "long-term debt-to-equity ratio"). Finally, analysts also use solvency ratios that deduct cash, cash equivalents, and marketable securities from interest-bearing short-term and long-term debt to calculate **net debt** (or **net cash** if the former exceeds the latter). The assumption is that cash, cash equivalents, and marketable securities could be used to pay debt obligations, so it is only debt in excess of this amount that must be covered by future operating cash flows. Analysts should be transparent about their calculation methodologies and closely scrutinize ratios reported by issuers.

Exhibit 25: Definitions of Commonly Used Solvency Ratios

Debt-to-assets ratio	Total debt	Total assets
Debt-to-capital ratio	Total debt[b]	Total debt[b] + Total shareholders' equity
Debt-to-equity ratio	Total debt[b]	Total shareholders' equity
Financial leverage ratio[c]	Average total assets	Average total equity
Debt-to-EBITDA	Total or net debt	EBITDA
Coverage Ratios		
Interest coverage	EBIT	Interest payments
Fixed charge coverage	EBIT + Lease payments	Interest payments + Lease payments

[a] "Total debt ratio" is another name sometimes used for this ratio.

[b] In this reading, total debt is the sum of interest-bearing short-term and long-term debt.

[c] Average total assets divided by average total equity is used for the purposes of this reading (in particular, Dupont analysis covered later). In practice, period-end total assets divided by period-end total equity is often used."

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Interpretation of Solvency Ratios

In the following, we discuss the interpretation of the basic solvency ratios presented in Exhibit 25.

Debt-to-Assets Ratio

This ratio measures the percentage of total assets financed with debt. For example, a **debt-to-assets ratio** of 0.40 or 40 percent indicates that 40 percent of the company's assets are financed with debt. Generally, higher debt means higher financial risk and thus weaker solvency.

Debt-to-Capital Ratio

The **debt-to-capital ratio** measures the percentage of a company's capital (debt plus equity) represented by debt. As with the previous ratio, a higher ratio generally means higher financial risk and thus indicates weaker solvency.

Debt-to-Equity Ratio

The **debt-to-equity ratio** measures the amount of debt capital relative to equity capital. Interpretation is similar to the preceding two ratios (i.e., a higher ratio indicates weaker solvency). A ratio of 1.0 would indicate equal amounts of debt and equity, which is equivalent to a debt-to-capital ratio of 50 percent. Alternative definitions of this ratio use the market value of stockholders' equity rather than its book value (or use the market values of both stockholders' equity and debt).

Financial Leverage Ratio

The **financial leverage ratio** (often called simply the "leverage ratio") measures the amount of total assets supported for each one money unit of equity. For example, a value of 3 for this ratio means that each EUR1 of equity supports EUR3 of total assets. The higher the financial leverage ratio, the more leveraged the company is in the sense of using debt and other liabilities to finance assets. This ratio is often defined in terms of average total assets and average total equity.

Debt-to-EBITDA Ratio

The debt-to-EBITDA ratio estimates how many years it would take to repay total debt based on earnings before income taxes, depreciation, and amortization (an approximation of operating cash flow). This ratio is commonly used in debt covenants between issuers and debt investors.

Interest Coverage

The **interest coverage ratio** measures the number of times a company's EBIT could cover its interest payments. Thus, it is sometimes referred to as "times interest earned." A higher **interest coverage** ratio indicates stronger solvency, offering greater assurance that the company can service its debt (i.e., bank debt, bonds, notes) from operating earnings. This ratio is commonly used in debt covenants between issuers and lenders or fixed income investors.

(before interest, taxes, and lease payments) can cover the company's interest and lease payments. Similar to the interest coverage ratio, a higher **fixed charge coverage** ratio implies stronger solvency, offering greater assurance that the company can service its debt (i.e., bank debt, bonds, notes, and leases) from normal earnings. The ratio is sometimes used as an indication of the quality of the preferred dividend, with a higher ratio indicating a more secure preferred dividend.

Example 11 demonstrates the use of solvency ratios in evaluating the creditworthiness of a company.

EXAMPLE 11

Evaluation of Solvency Ratios

A credit analyst is evaluating the solvency of Eskom, a South African public utility based on financial statements for the year ended 31 March 2017. The data in Exhibit 26 are gathered from the company's 2017 annual report:

Exhibit 26: Eskom 2017 Annual Report

South African rand, millions	2017	2016	2015
Total Assets	710,009	663,170	559,688
Short-Term Debt	18,530	15,688	19,976
Long-Term Debt	336,770	306,970	277,458
Total Liabilities	534,067	480,818	441,269
Total Equity	175,942	182,352	118,419

1. Calculate the company's financial leverage ratios for 2016 and 2017.

Hide Solution

Solution:

(Amounts are millions of Rand.)

For 2017, average total assets were (710,009 + 663,170)/2 = 686,590, and average total equity was (175,942 + 182,352)/2 = 179,147. Thus, financial leverage was 686,590/179,942 = 3.83. For 2016, financial leverage was 4.07.

	2017	2016
Average Assets	686,590	611,429
Average Equity	179,147	150,386
Financial Leverage	3.83	4.07

2. Interpret the financial leverage ratio calculated in question 1.

Hide Solution

Solution:

For 2017, every South African rand in total equity supported ZAR3.83 in total assets, on average. Financial leverage decreased from 2016 to 2017 on this measure.

3. What are the company's debt-to-assets, debt-to-capital, and debt-to-equity ratios for the three years?

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	2017	2016	2015
Total Debt	355,300	322,658	297,434
Total Capital	531,242	505,010	415,853
Debt/Assets	50.0%	48.7%	53.1%
Debt/Capital	66.9%	63.9%	71.5%
Debt/Equity	2.02	1.77	2.51

4. What is the discernable trend over the three years?

Hide Solution

Solution:

On all three metrics, the company's leverage decreased from 2015 to 2016 and increased from 2016 to 2017. For 2016 the decrease in leverage resulted from a conversion of subordinated debt into equity as well as additional issuance of equity. However, in 2017 debt levels increased again relative to assets, capital, and equity indicating that the company's solvency has weakened. From a creditor's perspective, lower solvency (higher debt) indicates higher risk of default on obligations.

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As with all ratio analysis, it is important to consider leverage ratios in a broader context. In general, companies with lower business risk and operations that generate steady cash flows are better positioned to take on more leverage without a commensurate increase in the risk of insolvency. In other words, a higher proportion of debt financing poses less risk of non-payment of interest and debt principal to a company with steady cash flows than to a company with volatile cash flows.

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GN

For purposes of the exam, what is the difference between debt-to-capital and debt-to-assets? Assuming A = L + E, and the assumption that 'total debt' includes only interest bearing debt, wouldn't the two be equal? Or is Assets book value and Capital is market value?

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For example 11

Debt-to-Assets 2015: 53.14% vs 2017: 50.04%, still 3.1pp below the 2015 figure

Debt-to-Capital 2015: 71.52% vs 2017: 66.88%, still 4.64pp below the 2015 figure

Debt-to-Equity 2015: 2.51x vs 2017: 2.02, still 0.49x below the 2015 figure

Assets grew year over year: 559,688 to 663,170 to 710,009

Debt grew as well, but at a slower pace: 297,434 to 322,658 to 355,300

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